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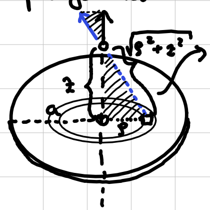
1 Električno polje nabite okrogle plošče



$\rightarrow \sigma$ (površinska gostota naboja)

a, σ

$E(z) = ?$

a) $z \ll a$

b) $z \gg a$

$$dE' = \frac{de}{4\pi\varepsilon_0\left(\sqrt{\rho^2+z^2}\right)^2}$$

$$\frac{dE}{dE'} = \frac{z}{\sqrt{\rho^2+z^2}}$$

$$dE = \frac{z\,dE'}{\sqrt{\rho^2+z^2}} = \frac{z}{\sqrt{\rho^2+z^2}}\,\frac{de}{4\pi\varepsilon_0\,(\rho^2+z^2)} = \frac{z\cdot\sigma\,2\pi\rho}{(\rho^2+z^2)^{3/2}\,2\pi\varepsilon_0}\,d\rho$$

$u = \rho^2+z^2$

$du = 2\rho\,d\rho$

$$E = \frac{z\sigma}{2\varepsilon_0}\int_0^a \frac{\rho}{(\rho^2+z^2)^{3/2}}\,d\rho = \frac{z\sigma}{2\varepsilon_0}\int_{z^2}^{a^2+z^2} u^{-3/2}\,\frac{du}{2} = \frac{z\sigma}{4\varepsilon_0}\,(-2)\,u^{-1/2}\Big|_{z^2}^{a^2+z^2}$$

$$= \frac{z\sigma}{2\varepsilon_0}\left(\frac{1}{z} - \frac{1}{\sqrt{a^2+z^2}}\right) = \boxed{\frac{\sigma}{2\varepsilon_0}\left(1-\frac{1}{\sqrt{\frac{a^2}{z^2}+1}}\right)}$$

a) $z \ll a$: $E = \frac{\sigma}{2\varepsilon_0}$ $\rightarrow$ el. polje zelo velike plošče

b) $z \gg a$: $\frac{a}{z} \ll 1$

$$\left(\left(\frac{a}{z}\right)^2+1\right)^{-1/2} = 1 - \frac{1}{2}\left(\frac{a}{z}\right)^2 \qquad (1+\varepsilon)^p = 1 + p\varepsilon + \binom{p}{2}\varepsilon^2 + \ldots$$

$\binom{p}{2} = \frac{p(p-1)}{2}$

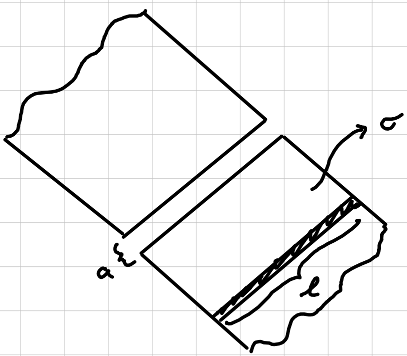
naboj na plošči

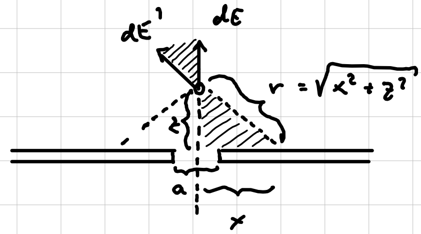
$$\rightarrow E = \frac{\sigma}{2\varepsilon_0}\left(\frac{1}{2}\left(\frac{a}{z}\right)^2\right) = \underbrace{\frac{\sigma}{4\varepsilon_0}\left(\frac{a}{z}\right)^2\cdot\frac{\pi}{\pi}}_{e} = \frac{e}{4\pi\varepsilon_0 z^2}$$

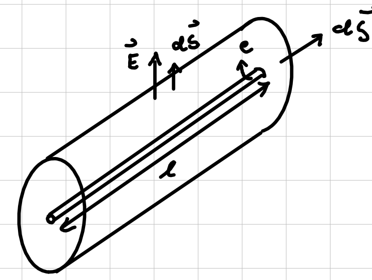
↳ točkast naboj

$\boxed{2}$ električno polje nabite ravne plošče z režo

a, σ







Podproblem: elek. polje nabitega ravnega vodnika:

↳ Gaussov izrek

Gauss: $$e = \varepsilon_0 \oint \vec{E}\cdot d\vec{S} = \varepsilon_0 \int_{\text{plašč}} E\, dS = \varepsilon_0 E S = \varepsilon_0 E \cdot 2\pi r l$$

$$E = \frac{e}{2\pi\varepsilon_0 l}\cdot\frac{1}{r}$$

$$de = l\,dx\,\sigma$$

$$\frac{dE}{dE'} = \frac{z}{(x^2+z^2)^{1/2}} \qquad dE' = \frac{de}{2\pi\varepsilon_0 l (x^2+z^2)} = \frac{dx\,\sigma}{2\pi\varepsilon_0 (x^2+z^2)^{1/2}}$$

→ dve polovici plošče

$$\int_0^E dE = \int_{a/2}^{\infty} \frac{z}{(x^2+z^2)^{1/2}}\,\frac{dx}{2\pi\varepsilon_0}\,\frac{\sigma\, 2}{(x^2+z^2)^{1/2}} = \frac{z\sigma}{\pi\varepsilon_0}\int_{a/2}^{\infty}(x^2+z^2)^{-1}dx = \frac{z\sigma}{\pi\varepsilon_0}\,\frac{1}{z}\,\operatorname{arctg}\frac{x}{z}\Big|_{a/2}^{\infty}$$

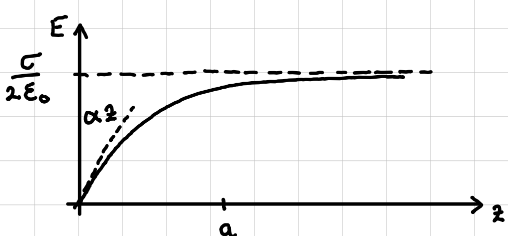
$$\boxed{E = \frac{\sigma}{\pi\varepsilon_0}\left(\frac{\pi}{2} - \operatorname{arctg}\frac{a}{2z}\right)}$$

Limite:

a) $z \gg a \Rightarrow E(z) \approx \frac{\sigma}{2\varepsilon_0}$ → neskončna ravna plošča (kot da ne bi bilo reže) – smiselno

b) $z \ll a$: $\operatorname{arctg} x = \frac{\pi}{2} - x + \dots$ razvoj za $x \ll 1$

$$E(z) \approx \frac{\sigma}{\pi\varepsilon_0}\left(\frac{\pi}{2} - \frac{\pi}{2} + \frac{2z}{a}\right) = \frac{2\sigma z}{\pi\varepsilon_0 a} = \frac{2\sigma}{\pi\varepsilon_0 a}\cdot z$$



3 Poissonova enačba za točkast naboj

$$\vec{\nabla}\cdot\vec{E}(\vec{r}) = \frac{\rho(\vec{r})}{\varepsilon_0} \rightarrow \text{prostorninska gostota naboja}$$

$$\vec{E}(\vec{r}) = -\vec{\nabla}U(\vec{r}) \ , \quad \boxed{\nabla^2 U = -\frac{\rho(\vec{r})}{\varepsilon_0}} \quad \text{Poissonova enačba}$$

$$\rho(\vec{r}) = e\underbrace{\delta(\vec{r}-\vartheta)}_{1/m^3} \qquad \boxed{\nabla^2 U(\vec{r}) = -\frac{e}{\varepsilon_0}\delta(\vec{r})}$$

FT slovarček:

$\vec{\nabla} \xrightarrow{FT} i\vec{k}$

$\nabla^2 \xrightarrow{FT} -k^2$

Tehnični del:

Fourierova transformacija:

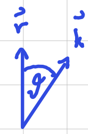
$$\boxed{U(\vec{r}) = \underbrace{\int d^3\vec{k}}_{\text{"vsota"}}\ \underbrace{U(\vec{k})}_{\text{amplitude (uteži) ravnih valov}}\ \underbrace{e^{i\vec{k}\vec{r}}}_{\text{ravni val } \vec{k}}} \quad \Big/ \overbrace{\int d^3\vec{r}\, e^{-i\vec{k}'\vec{r}}}^{\text{skalarni produkt z drugo bazno funkcijo}}$$

$$\int d^3\vec{r}\, U(\vec{r})\, e^{-i\vec{k}'\vec{r}} = \iint d^3\vec{r}\, d^3\vec{k}\, U(\vec{k})\, \underbrace{e^{i(\vec{k}-\vec{k}')\vec{r}}}_{(2\pi)^3\delta(\vec{k}-\vec{k}')} = \int d^3\vec{k}\,(2\pi)^3\delta(\vec{k}-\vec{k}')\,U(\vec{k}) = (2\pi)^3 U(\vec{k}')$$

$$\boxed{U(\vec{k}) = \frac{1}{(2\pi)^3}\int d^3\vec{r}\, U(\vec{r})\, e^{-i\vec{k}\vec{r}}} \quad \text{izraz za amplitude} \longrightarrow \frac{1}{(2\pi)^3}\underbrace{\int d^3\vec{r}\,\delta(\vec{r}-0)\,e^{-i\vec{k}\vec{r}}}_{1} = \frac{1}{(2\pi)^3}$$

$$\vec{\nabla}U(\vec{r}) = \int d^3\vec{k}\, U(\vec{k})\underbrace{\frac{\partial}{\partial\vec{r}}e^{i\vec{k}\vec{r}}}_{i\vec{k}e^{i\vec{k}\vec{r}}} = \int d^3\vec{k}\left[i\vec{k}\,U(\vec{k})\right]e^{i\vec{k}\vec{r}}$$

$\hookrightarrow \frac{\partial}{\partial\vec{r}}$



$$\nabla^2 U(\vec{r}) = -\frac{e}{\varepsilon_0}\delta^3(\vec{r})$$

posamezne amplitude

$$-k^2 U(\vec{k}) = -\frac{e}{\varepsilon_0}\frac{1}{(2\pi)^3} \rightarrow u(\vec{k}) = -\frac{e}{(2\pi)^3\varepsilon_0}\frac{1}{k^2}$$

$d^3\vec{k} = k^2\, d(\cos\vartheta)\, dk\, d\varphi$

ne smemo najprej int. po k, ker dobimo ∞

$$u(\vec{r}) = \int d^3k\, u(\vec{k})e^{i\vec{k}\vec{r}} = \frac{e}{(2\pi)^3\varepsilon_0}\int d^3k\,\frac{e^{i\vec{k}\vec{r}}}{k^2} = \frac{e\cdot \overbrace{2\pi}^{\int d\varphi}}{(2\pi)^3\varepsilon_0}\int_{-1}^{1}d(\cos\vartheta)\int_0^\infty k^2 dk\,\frac{e^{ikr\cos\vartheta}}{k^2} =$$

$$= \frac{e}{(2\pi)^3\varepsilon_0}\int_0^\infty dk\,\frac{1}{irk}\left(e^{ikr}-e^{-ikr}\right) = \frac{2e}{(2\pi)^2\varepsilon_0 r}\int_0^\infty d(kr)\,\frac{1}{kr}\sin(kr) = \qquad kr = u$$

$$= \frac{2e}{(2\pi)^2\varepsilon_0 r}\underbrace{\int_0^\infty du\,\frac{\sin u}{u}}_{\frac{\pi}{2}} = \boxed{\frac{e}{4\pi\varepsilon_0 r}}$$

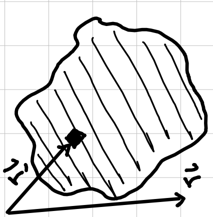
$$\nabla^2\frac{e}{4\pi\varepsilon_0 r} = -\frac{e}{\varepsilon_0}\delta(\vec{r}) \ , \quad \nabla^2\frac{1}{r} = -4\pi\delta(\vec{r})$$

točkasti naboj:

$$\rho(\vec{r}) = e\delta(\vec{r}) \xrightarrow{\text{Poissonova enačba:}} U(\vec{r}) = \frac{e}{4\pi\varepsilon_0|\vec{r}-0|}$$

Poljubna porazdelitev sestavimo iz točkastih nabojev:

$$\rho(\vec{r}) = \int d^3\vec{r}'\,\rho(\vec{r}')\,\delta(\vec{r}-\vec{r}') \longrightarrow U(\vec{r}) = \int\frac{\overbrace{d^3\vec{r}'\,\rho(\vec{r}')}^{de}}{4\pi\varepsilon_0|\vec{r}-\vec{r}'|}$$



$$\vec{E}(\vec{r}) = -\vec{\nabla}U(\vec{r}) = -\frac{\partial}{\partial\vec{r}}U(\vec{r})$$

$$\boxed{\vec{E}(\vec{r}) = \int\frac{\overbrace{d^3\vec{r}'\,\rho(\vec{r}')}^{de}}{4\pi\varepsilon_0|\vec{r}-\vec{r}'|^2}\,\frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|}}$$

4 gostota nabgja v vodikovem atomu (v osnovnem stanju)

sfemo simetrično

$$U(r) = \frac{e}{4\pi\varepsilon_0} \frac{e^{-\alpha r}}{r}\left(1 + \frac{\alpha r}{2}\right), \quad \alpha = \frac{2}{r_B} \leftarrow \text{Bohrov radij}$$

$$\rho(r) = ?$$

$$\xrightarrow{r\to 0} \frac{e}{4\pi\varepsilon_0 r}$$

vemo, da je en prispevek, ki je δ fun.

singularnost v $r=0$ nam da 2 prispevka

$$\rightarrow \rho_+(\vec{r}) = e\delta(\vec{r})$$

sferna simetrija:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right), \quad \nabla^2 u = -\frac{\rho(\vec{r})}{\varepsilon_0} = \nabla^2\left(\frac{e}{4\pi\varepsilon_0}\frac{e^{-\alpha r}}{r}\left(1+\frac{\alpha r}{2}\right)\right) = \frac{e}{4\pi\varepsilon_0}\nabla^2\left(\frac{e^{-\alpha r}}{r} + \frac{\alpha}{2}e^{-\alpha r}\right) =$$

$$= \frac{e}{4\pi\varepsilon_0}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\left(\frac{-\alpha e^{-\alpha r} r - e^{-\alpha r}}{r^2} - \frac{\alpha^2}{2}e^{-\alpha r}\right)\right) = \frac{e}{4\pi\varepsilon_0}\frac{1}{r^2}\frac{\partial}{\partial r}\left(-\alpha e^{-\alpha r} r - e^{-\alpha r} - \frac{\alpha^2 r^2}{2}e^{-\alpha r}\right) =$$

$$= -\frac{e}{4\pi\varepsilon_0}\frac{1}{r^2}\frac{\partial}{\partial r}\left(e^{-\alpha r}\left(1 + \alpha r + \frac{\alpha^2 r^2}{2}\right)\right) = -\frac{e}{4\pi\varepsilon_0 r^2}\left(-\alpha e^{-\alpha r}\left(1 + \alpha r + \frac{\alpha r^2}{2}\right) + e^{-\alpha r}(\alpha + \alpha^2 r)\right) =$$

$$= -\frac{e}{4\pi\varepsilon_0 r^2}\left(e^{-\alpha r}\left(-\alpha - \alpha^2 r + \frac{\alpha^3 r^2}{2} + \alpha + \alpha^2 r\right)\right) = \frac{e}{4\pi\varepsilon_0 r^2}\cdot\frac{e^{-\alpha r}\alpha^3 r^2}{2} = \frac{e\, e^{-\alpha r}\alpha^3}{8\pi\varepsilon_0}$$

$$\rho(r) = -\frac{e e^{-\alpha r}\alpha^3}{8\pi} \rightarrow \boxed{\rho(r) = -\frac{e\alpha^3}{8\pi}e^{-\alpha r}} \rightarrow \text{elektron}$$

→ singularnost $U(r)$ v $r=0$ nam da 2 prispevka

$$\rho(\vec{r}) = \rho_+(\vec{r}) + \rho_-(\vec{r}) \Rightarrow \boxed{\rho(\vec{r}) = e\delta(\vec{r}) - \frac{e\alpha^2}{8\pi}e^{-\alpha r}}$$

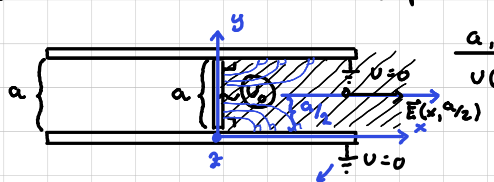
razen v kakšnih točkah

Če $\rho(\vec{r}) = 0$ skoraj povsod $\Rightarrow$ $\boxed{\nabla^2 U(\vec{r}) = 0}$ skoraj povsod

Laplaceova enačba

tam kjer ni : $\boxed{\text{robni pogoji}}$

$\boxed{5}$ Ploščni kondenzator s prečnim trakom



$\underline{a, u_0}$

$U(x, y, \cancel{z}) = 0$ znotraj kondenzatorja

↳ vzdolž z je vse enako

od tu lahko priteče naboj (zemlja) ima velike zaloge naboja

$$\nabla^2 U(x,y) = 0 \quad , \quad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

Robni pogoji:

1.) spodnja plošča : $u(x, 0) = 0$ — ne moremo dobiti ∞, ker je vse končno

2.) zgornja plošča : $u(x, a) = 0$

3.) trak : $u(0, y) = u_0$ $\Big\} \; 0 < y < a$

4.) desno : $u(\infty, y) \neq \infty$

Separacija spremenljivk : $U(x,y) = X(x)\, Y(y)$

$$X''Y + XY'' = 0 \quad / \frac{1}{XY}$$

$$\frac{X''}{X} + \frac{Y''}{Y} = 0 \quad \rightarrow \quad \frac{X''}{X} = -\frac{Y''}{Y} = \varkappa^2 \qquad \forall\, x, y \text{ v našem območju}$$

$$X'' - \varkappa^2 X = 0 \quad \longrightarrow \quad X = \overset{0 \text{ (RP4)}}{\cancel{C}} e^{\varkappa x} + D e^{-\varkappa x}$$

$$Y'' + \varkappa^2 Y = 0 \quad \longrightarrow \quad Y = A \sin \varkappa y + B \cos \varkappa y$$

$$U(x,y) = (A \sin \varkappa y + B \underset{\text{RP1}}{\cancel{\cos}} \varkappa y)(\cancel{C e^{\varkappa x}} + D e^{-\varkappa x})$$

$$U(x, 0) = \underset{0}{B} \cdot D e^{-\varkappa x} = 0$$

$$U(x,y) = \underset{AD}{F} \sin \varkappa y \cdot e^{-\varkappa x}$$

$$0 = F \cdot \underbrace{\sin(\varkappa a)}_{0} e^{-\varkappa x}$$

$$a\varkappa = n\pi \Rightarrow \varkappa = \frac{n\pi}{a}, \quad n = 1, 2, 3, \ldots$$

sk. produkt z drugo bazno fun.

$$U(x,y) = \sum_{n=1}^{\infty} F_n \sin\left(\frac{n\pi}{a} y\right) e^{-\frac{n\pi}{a} x}$$

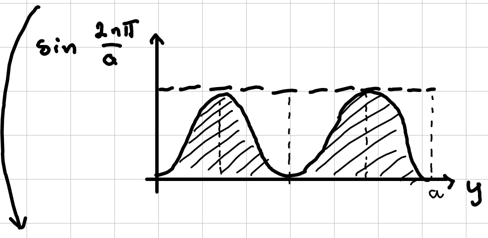
RP 3: $$U(0, 0 < y < a) = \sum_{n=1}^{\infty} F_n \sin\left(\frac{n\pi}{a} y\right) \cdot 1 = U_0 \quad / \int_0^a dy \sin\left(\frac{m\pi}{a} y\right)$$

sk. produkt dveh baznih fun.

↳ so vsaj ortogonalne

leva: $$\int_0^a \sum_{n=1}^{\infty} F_n \sin\left(\frac{n\pi}{a} y\right) \cdot \sin\left(\frac{m\pi}{a} y\right) dy = \sum_{n=1}^{\infty} \underbrace{\int_0^a F_n \sin\left(\frac{n\pi}{a} y\right) \cdot \sin\left(\frac{m\pi}{a} y\right) dy}_{\delta_{mn}}$$

$$\underbrace{\int_0^a \sin^2\left(\frac{2\pi}{a} y\right) dy}_{\frac{1}{2} \cdot a} \rightarrow \text{dolžina intervala}$$



desno stran

$$= \sum_{n=1}^{\infty} \frac{F_n a}{2} \delta_{mn} = \boxed{\frac{F_m a}{2}}$$

D.S.: $$\int_0^a U_0 \cdot \sin\left(\frac{n\pi}{a} y\right) dy = U_0 \left. \frac{-\cos\left(\frac{n\pi}{a} y\right) \cdot a}{m\pi} \right|_0^a = \frac{U_0 a}{m\pi}\left(\cos m\pi - 1\right) = \underline{\frac{U_0 a}{m\pi}\left(1 - (-1)^m\right)}$$

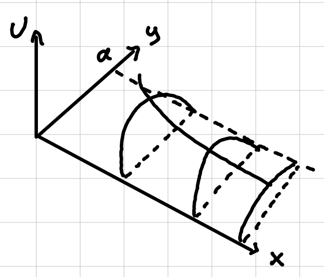
$$\Rightarrow \quad \frac{F_m a}{2} = -\left((-1)^m - 1\right) \frac{U_0 a}{m\pi}$$

$$F_m = \left(1 - (-1)^m\right) \frac{2U_0}{m\pi}$$

$$U(x,y) = \sum_{m=1}^{\infty} \frac{2U_0}{m\pi}\left(1 - (-1)^m\right) \sin\left(\frac{m\pi}{a} \cdot y\right) \cdot e^{-\frac{m\pi}{a} x}$$

a) $x \gg a \rightarrow$ preživi le $m = 1$

$$U(x,y) = \frac{4U_0}{\pi} \sin\left(\frac{\pi}{a} y\right) e^{-\frac{\pi}{a} x}$$



b) $\vec{E}(x, a/2) \longrightarrow \overset{\text{samo}}{E_x}$

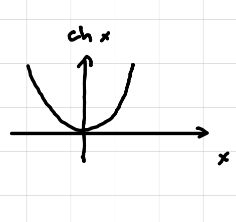
m	1	2	3	4	5	6	7
	2	0	-2	0	2	0	-2

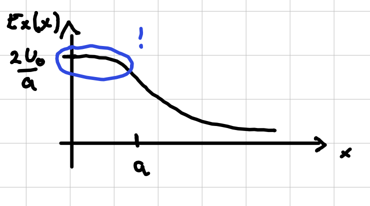
$$E_x(x) = -\frac{\partial U(x, a/2)}{\partial x} = -\frac{\partial}{\partial x} \sum_m \frac{2U_0}{m\pi}\left(1 - (-1)^m\right) \sin\left(\frac{m\pi}{a} \frac{a}{2}\right) e^{-\frac{m\pi}{a} x} =$$

$$= \sum \frac{2U_0}{m\pi}\left(1 - (-1)^m\right) \sin\left(\frac{m\pi}{2}\right)\left(+\frac{m\pi}{a}\right) e^{-\frac{m\pi}{a} x} = \frac{2U_0}{a} \sum_{m=1}^{\infty} \overbrace{\left(1 - (-1)^m\right) \sin\left(\frac{m\pi}{2}\right)} e^{-\frac{m\pi}{a} x} =$$

$$= \frac{4U_0}{a}\left[e^{-\frac{\pi x}{a}} - e^{-\frac{3\pi x}{a}} + e^{-\frac{5\pi x}{a}} - e^{-\frac{7\pi x}{a}} \ldots\right] = \frac{4U_0}{a} e^{-\frac{\pi x}{a}} \underbrace{\left[1 - \underset{\alpha}{e^{-\frac{2\pi x}{a}}} + e^{-\frac{4\pi x}{a}} \ldots\right]}_{1 - \alpha + \alpha^2 - \alpha^3 + \ldots = \frac{1}{1+\alpha}}$$

$$= \frac{4U_0}{a} \frac{1}{e^{\frac{\pi x}{a}} + e^{-\frac{\pi x}{a}}} = \boxed{\frac{2U_0}{a \operatorname{ch}\left(\frac{\pi x}{a}\right)}}$$



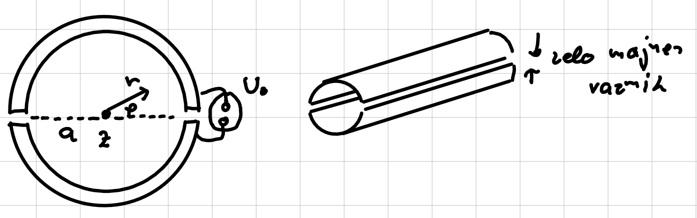


6 <u>prepolovljena prevodna cev</u>

a, U_0

$U(r,\varphi) = ?$

znotraj



$$\nabla^2 U(r,\varphi) = 0 \quad , \qquad \text{separacija: } U(r,\varphi) = R(r)\,\Phi(\varphi)$$

$$\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \varphi^2}\right] R\Phi = 0$$

$$\frac{1}{r}(rR')'\,\Phi + \frac{1}{r^2} R\Phi'' = 0$$

$$\frac{1}{r}(R' + rR'')\Phi + \frac{R}{r^2}\Phi'' = 0 \quad \Big/ \frac{1}{\frac{R}{r^2}\Phi}$$

$$\frac{\frac{1}{r}R' + R''}{R/r^2} = -\frac{\Phi''}{\Phi}$$

$$\frac{rR' + r^2R''}{R} = -\frac{\Phi''}{\Phi} = \underbrace{m^2}_{\text{konstanta}} \quad , \quad \forall\, r, \varphi$$

$$\Phi'' + m^2\Phi = 0 \longrightarrow \Phi(\varphi) = A_m \sin(m\varphi) + B_m\cos(m\varphi)$$

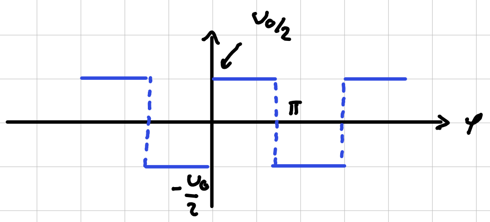
$$r^2R'' + rR' - m^2R = 0 \longrightarrow R(r) = C_m r^m + D_m r^{-m}$$

splošna reš.: $$U(r,\varphi) = \sum_{m=1}^{\infty}\left(A_m\sin(m\varphi) + B_m\cos(m\varphi)\right)\left(C_m r^m + D_m r^{-m}\right)$$

to je na kol. in izpitih podano

<u>Robni pogoji</u>:

1 $$U(a,\varphi) = \begin{cases} U_0/2, & 0 < \varphi < \pi \\ -U_0/2, & \pi < \varphi < 2\pi \end{cases}$$



2 $U(0,\varphi) \neq \infty \Rightarrow \boxed{D_m = 0}$

$$U(r,\varphi) = \sum_{m=1}^{\infty} \overset{A_m C_m}{F_m} \sin m\varphi \cdot r^m$$

$$\left.\begin{matrix} U_0/2 \\ -U_0/2 \end{matrix}\right\} \sum_{m=1}^{\infty} F_m a^m \sin(m\varphi) \quad \Big/ \int_0^{2\pi} d\varphi\, \sin(n\varphi)$$

desna: $$\int_0^{2\pi} d\varphi \sum_{m=1}^{\infty} F_m a^m \sin(m\varphi)\sin(n\varphi) = \sum_{m=1}^{\infty} a^m F_m \int_0^{2\pi} \sin(\varphi m)\sin(\varphi n)\, d\varphi =$$

$$= \sum_{m=1}^{\infty} a^m F_m \delta_{mn} \underbrace{\int_0^{2\pi} d\varphi \sin^2(m\varphi)}_{\frac{1}{2}(2\pi)} = \sum_{m=1}^{\infty} a^m F_m \delta_{mn}\pi = \boxed{a^n F_n \pi}$$

leva: $$\int_0^{\pi} d\varphi \sin(n\varphi)\frac{U_0}{2} - \int_{\pi}^{2\pi} d\varphi \sin(n\varphi)\frac{U_0}{2} = \frac{U_0}{2}\left(-\frac{\overset{(-1)^n}{\cos(n\varphi)}}{n}\Big|_0^{\pi} + \frac{\cos(n\varphi)}{n}\Big|_{\pi}^{2\pi}\right) = \frac{U}{2n}\left[(1-(-1)^n) + (1-(-1)^n)\right] =$$

$$= \boxed{\frac{U_0}{n}\left(1-(-1)^n\right)}$$

$$a^n F_m \pi = \frac{U_0}{n}\left(1-(-1)^n\right)$$

$$F_m = \frac{U_0\left(1-(-1)^n\right)}{n\pi a^n}$$

$$\boxed{U(r,\varphi) = \sum_{m=1}^{\infty} \frac{U_0}{m\pi}\left(1-(-1)^m\right)\sin(m\varphi)\left(\frac{r}{a}\right)^m}$$